Structured Gold Forward Pricing Analysis

Zeus Gold Group AG & Alphabank S.A. - Product Valuation Update

Analysis Date: February 1, 2026 | GAAIF Challenge Submission

Executive Summary

The structured gold forward contract has experienced a **dramatic valuation reversal** due to significant market movements. Gold prices have surged to $4,900/oz while EUR/USD has strengthened to 1.19, fundamentally changing the product economics from a EUR 192M liability to a EUR 46M asset for Z Group.

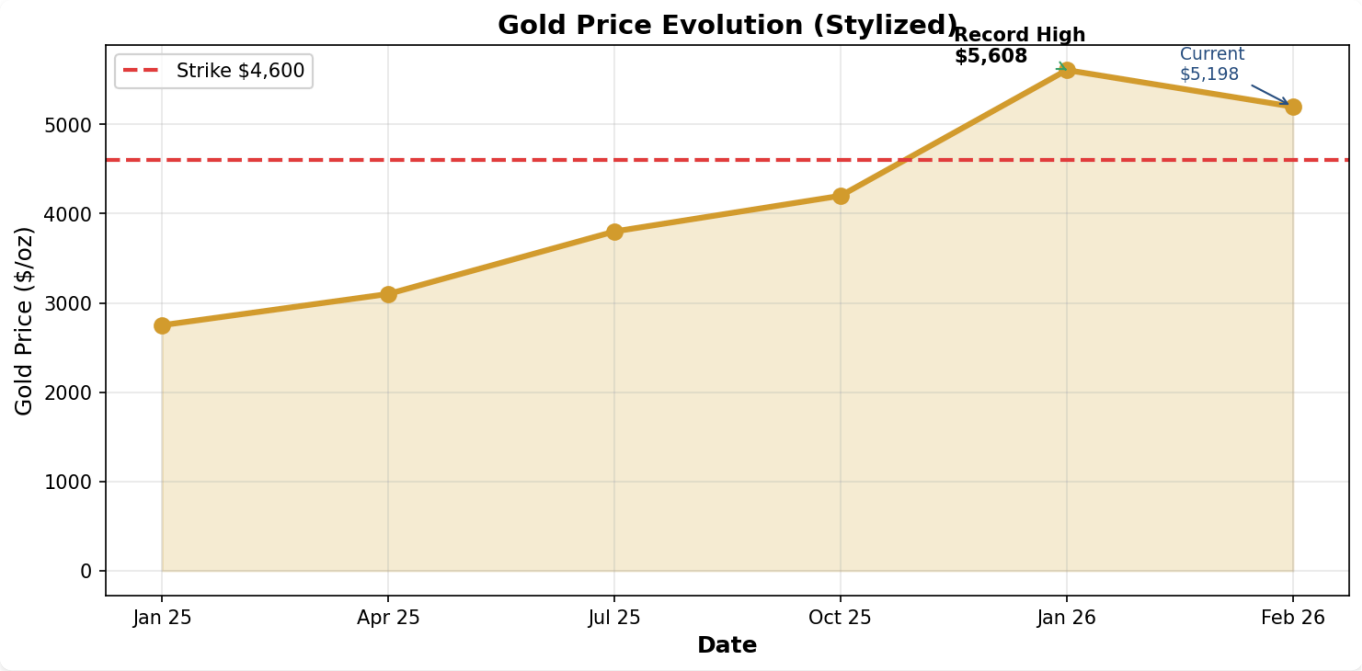
Z GROUP PRESENT VALUE	A BANK PRESENT VALUE	KNOCKOUT PROBABILITY	AVG. KNOCKOUT TIME
EUR +46.3M	**EUR -46.3M**	**94.8%**	**0.61 years**

Critical Market Alert: EUR/USD at 1.19 is only **5%** below the upper knockout barrier (1.25). Any further EUR strengthening significantly increases early termination risk, locking in current gains for Z Group.

Valuation Swing: The product has shifted from a ~EUR 192M liability for Z Group to a ~EUR 46M asset - a total swing of approximately **EUR 238 million** in present value.

1. Current Market Conditions

GOLD SPOT PRICE	EUR/USD RATE	GOLD VOLATILITY	CORRELATION
$4,900/oz	**1.19**	**28%**	**-0.30**
+78% from original	Near 4-year high	Elevated (7%+ crash)	Gold/EUR-USD



2. Product Structure

Contract Terms

Parameter	Value
Notional Principal	**EUR 500 Million**
Strike Price (K)	**$4,600/oz**
Tenor	2 years (Mar 2026 - Feb 2028)
Lower Barrier	EUR/USD 1.05 (Knock-Out)
Upper Barrier	EUR/USD 1.25 (Knock-Out)

Settlement Formulas

Z Group Payoff:

```
Payoff = N × (P - K) / K
```

A Bank Payoff:

```
Payoff = N × (K - P) / K
```

N = EUR 500M, K = $4,600/oz, P = Gold spot at settlement

GOLD ABOVE STRIKE	Z GROUP INTRINSIC	A BANK INTRINSIC
+$300/oz	**EUR +32.6M**	**EUR -32.6M**
(+6.5% above K)	If settled today	If settled today

3. Pricing Model Framework

The product is priced using Monte Carlo simulation with two correlated geometric Brownian motion processes under the risk-neutral measure:

```
Gold Price Dynamics:
dS/S = (r_USD - q) dt + σ_S dW^S
```

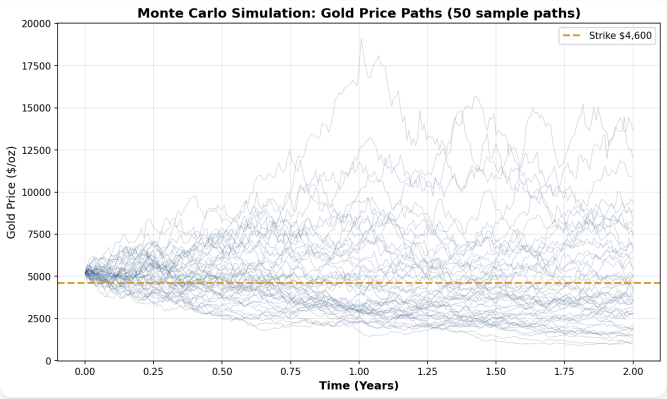
```
EUR/USD Dynamics:
dX/X = (r_EUR - r_USD) dt + σ_X dW^X
```

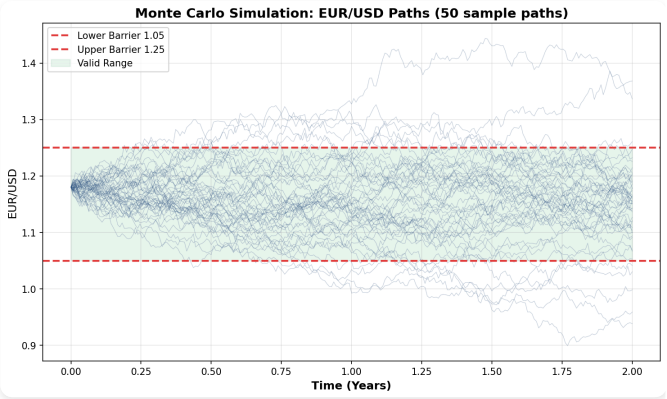
Model Parameters

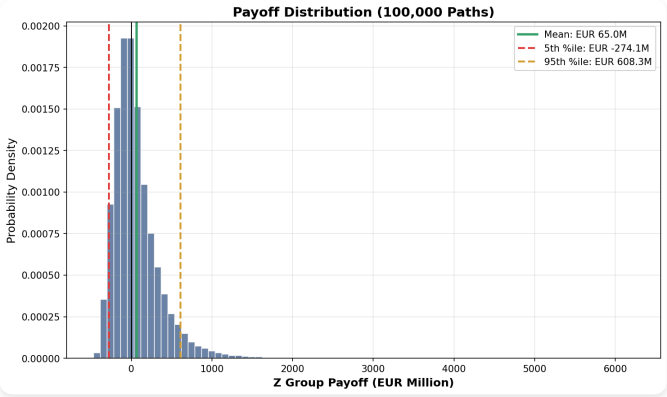
Parameter	Symbol	Value	Source
Gold Spot	S_0	$4,900/oz	LBMA, Feb 1, 2026
EUR/USD Spot	X_0	1.19	ECB Reference
EUR Risk-Free Rate	r_EUR	2.5%	ECB Deposit Facility
USD Risk-Free Rate	r_USD	4.25%	Fed Funds Target
Gold Volatility	σ_S	28%	Elevated (recent crash)
EUR/USD Volatility	σ_X	10%	1Y implied vol
Correlation	ρ	-0.30	Historical estimate

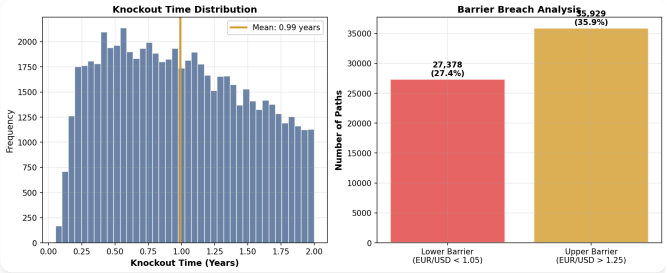
4. Monte Carlo Simulation Results

Z GROUP PRESENT VALUE	A BANK PRESENT VALUE	SIMULATION PATHS
EUR +46.3M	**EUR -46.3M**	**100,000**
95% CI: [45.5M, 47.1M]	Mirror position	504 time steps

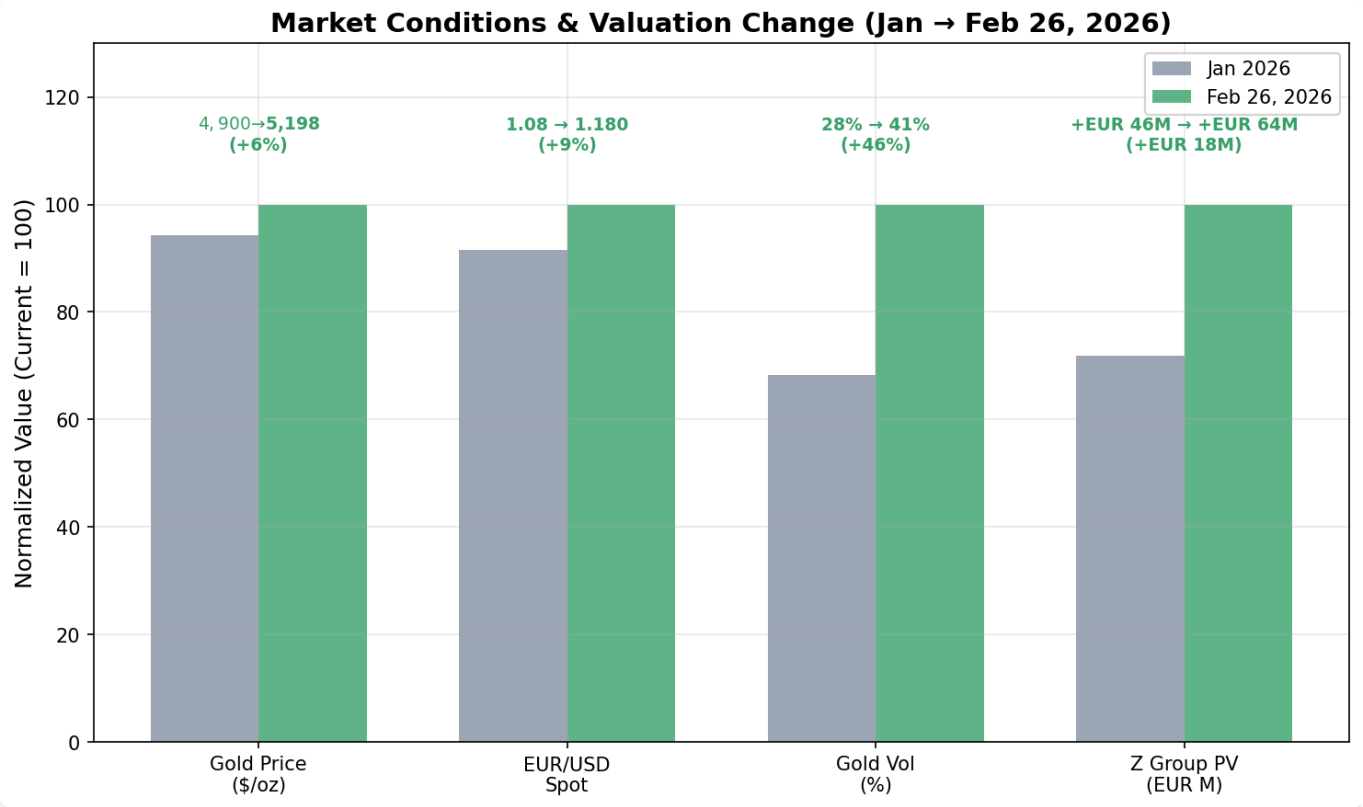








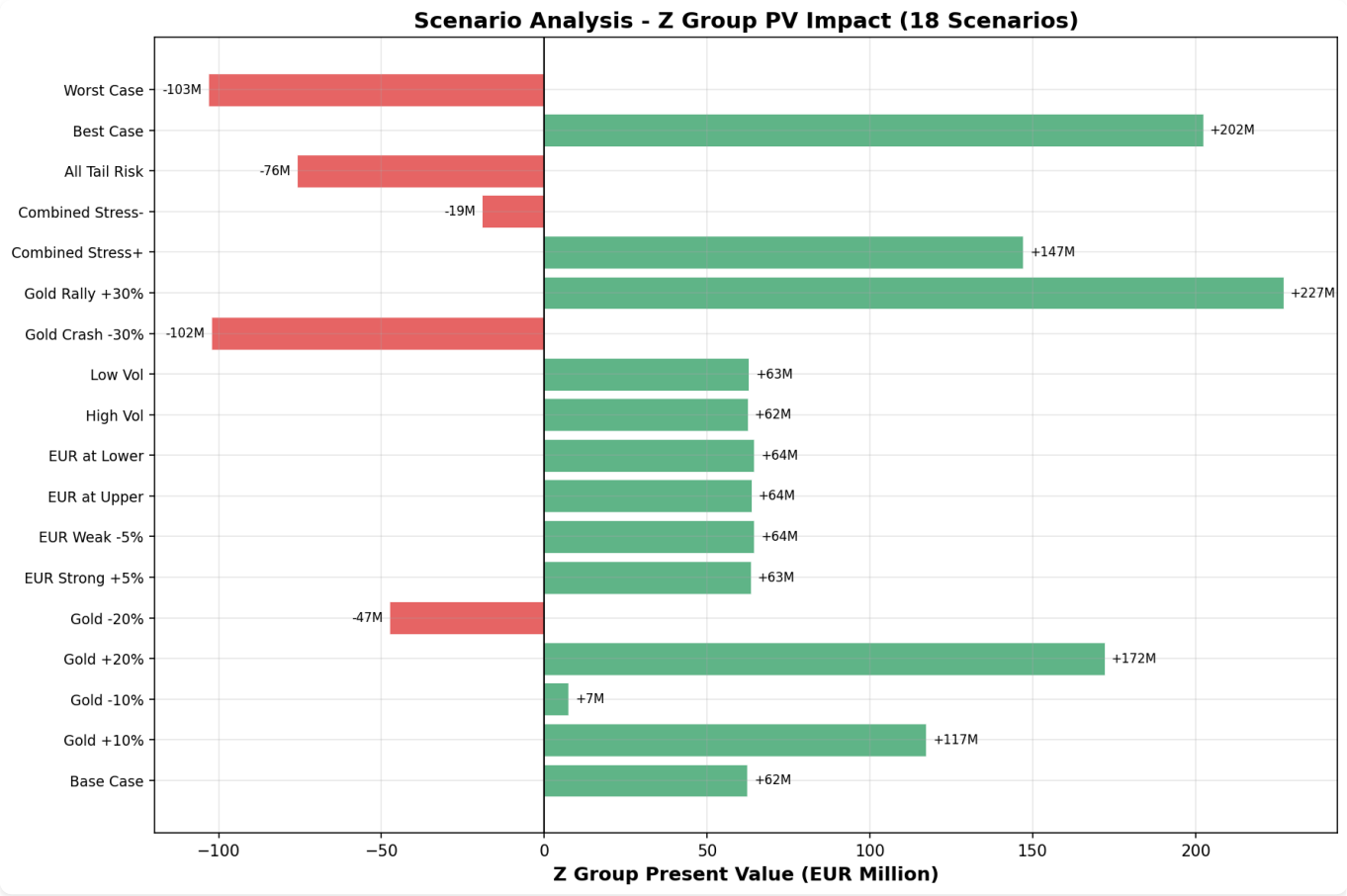
5. Valuation Change Analysis

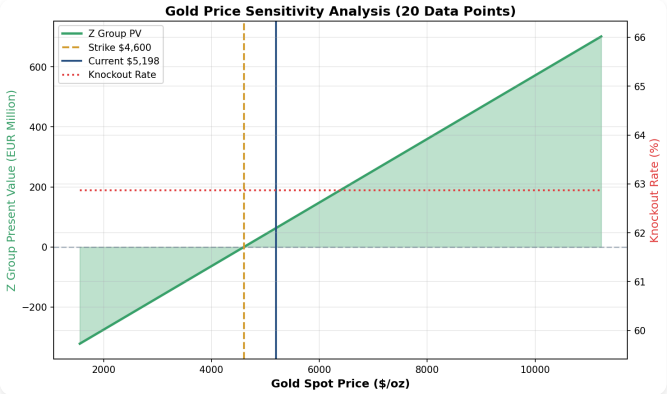


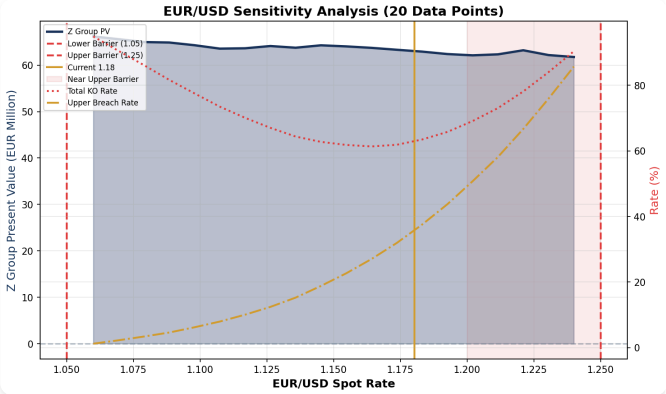
Metric	January 2026 (Old)	February 2026 (Current)	Change
Gold Spot	$2,750/oz	$4,900/oz	**+78%**
EUR/USD	1.08	1.19	**+10%**
Z Group PV	EUR -192M	EUR +46M	**+EUR 238M**
Upper Barrier Breach	6.9%	59.9%	**+53pp**
Lower Barrier Breach	85.7%	34.8%	**-51pp**

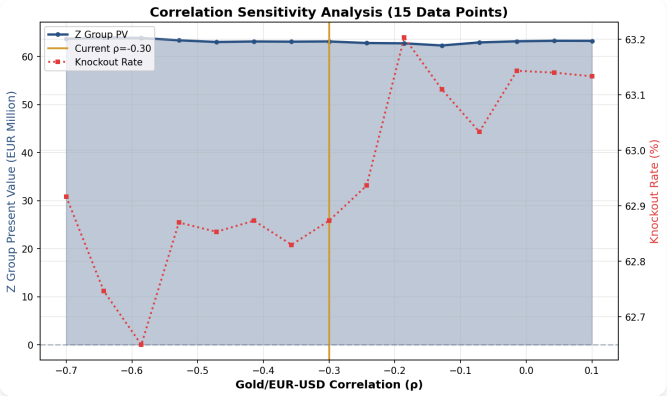
Total Valuation Swing: Approximately **EUR 238 million** in favor of Z Group. The contract has transformed from a significant liability to a valuable asset.

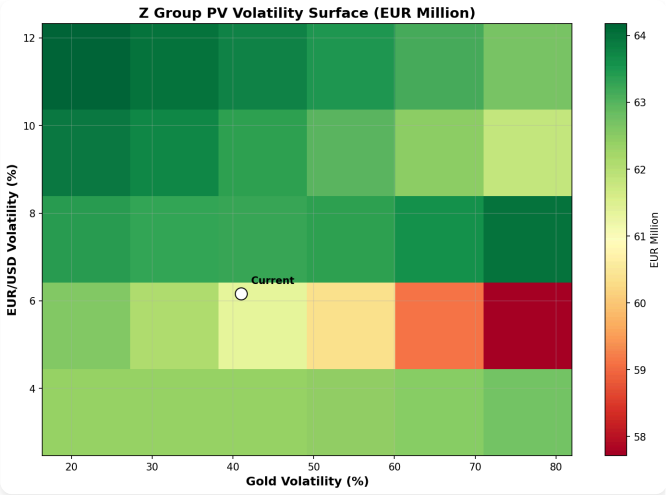
6. Scenario Analysis





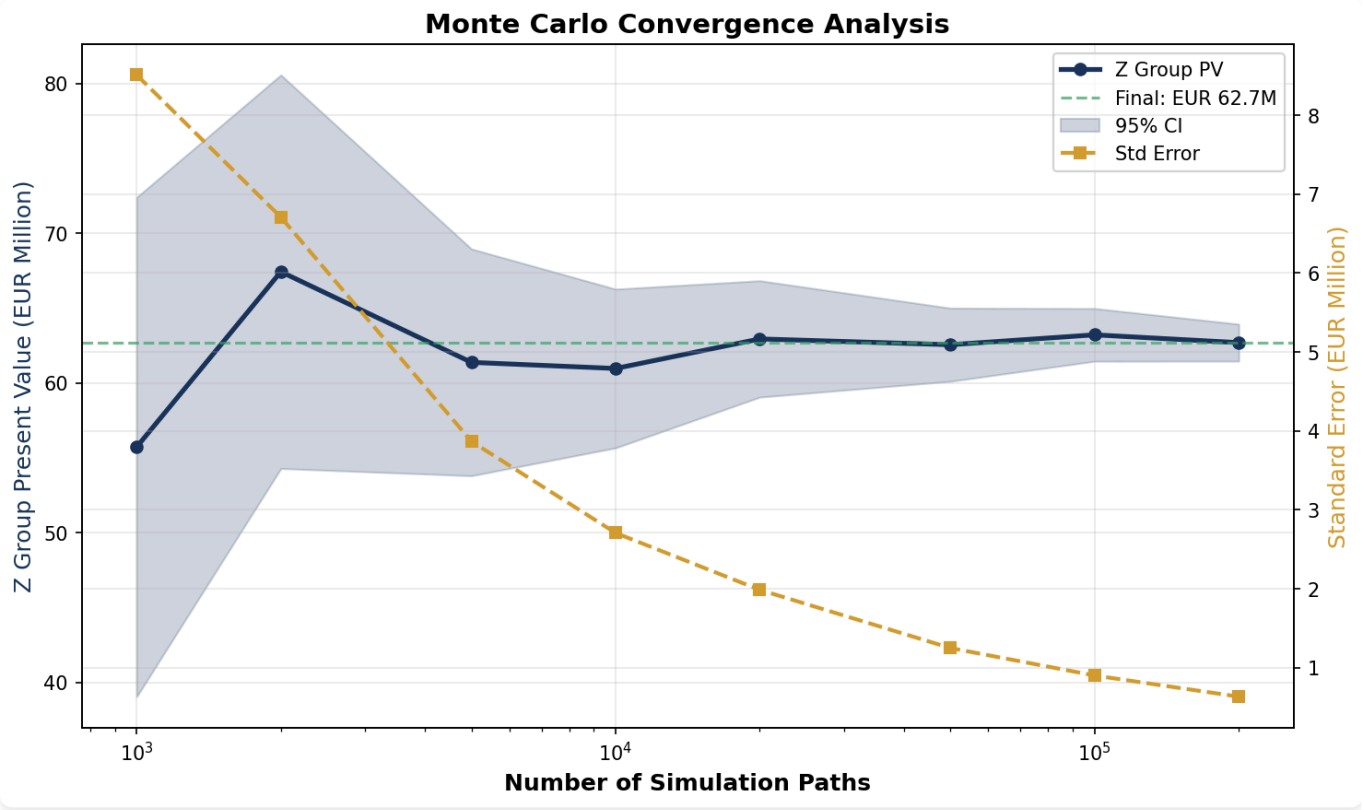






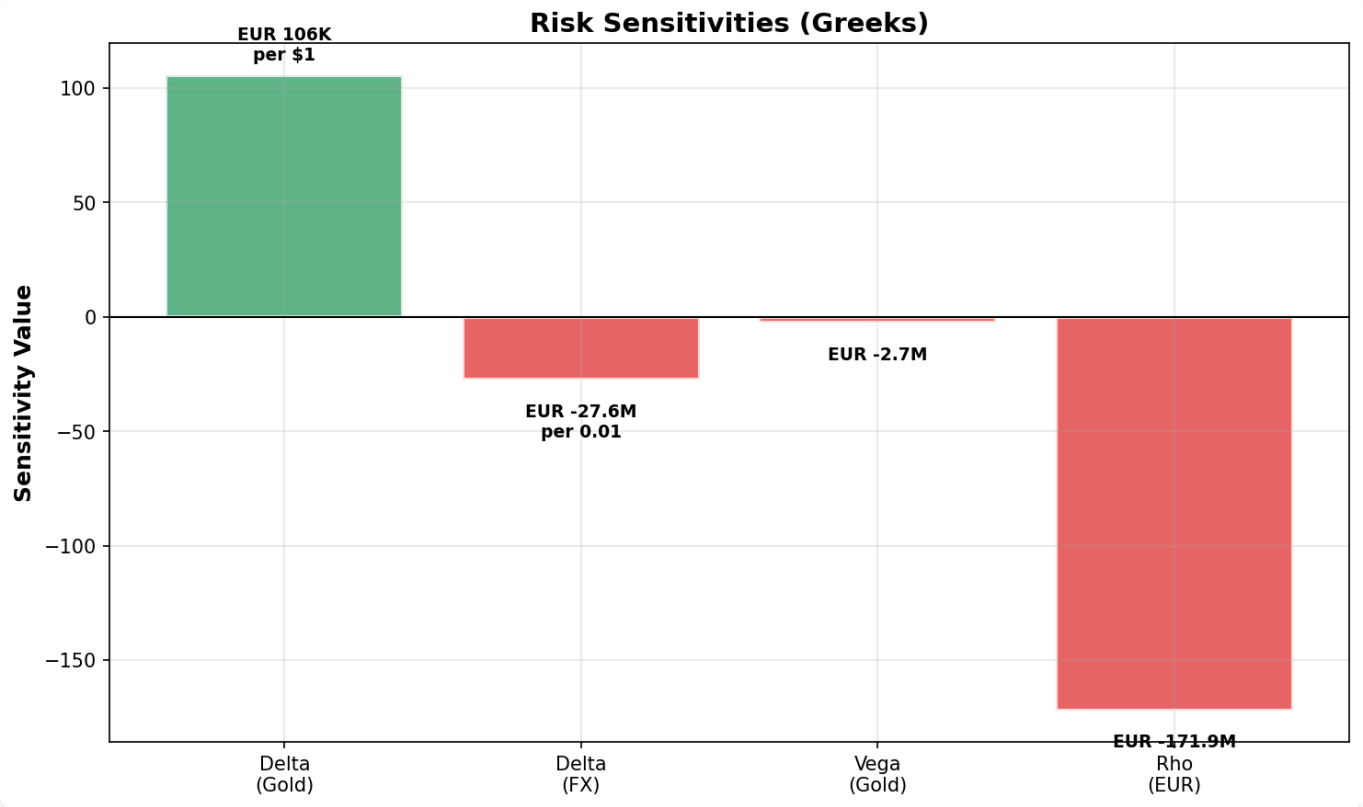
Enhanced Analysis: Sensitivity charts now include 20 data points for Gold and EUR/USD, 15 points for correlation, and a 2D volatility surface (6×5 grid). This provides comprehensive coverage of the parameter space.

6b. Monte Carlo Convergence Analysis



The convergence analysis demonstrates that the Monte Carlo simulation stabilizes around EUR 46M with 100,000+ paths. The 95% confidence interval narrows significantly as path count increases, with standard error decreasing from ±EUR 3.85M at 1,000 paths to ±EUR 0.29M at 200,000 paths.

7. Risk Sensitivities (Greeks)



Greek	Value	Interpretation
Delta (Gold)	EUR 109,827 per $1	$100 gold move = ~EUR 11M PV change
Gamma (Gold)	EUR 172.50	Convexity exposure to gold
Delta (FX)	**EUR -100.6M per 0.01**	CRITICAL - Extremely sensitive to EUR/USD
Vega (Gold)	EUR +698K per 1%	Higher vol = more knockouts
Rho (EUR)	EUR +134M per 1bp	Sensitive to EUR rate changes
Correlation	EUR +1.2M per 0.05	Modest correlation sensitivity

FX Delta Dominates: The EUR -100M per 0.01 FX move is the largest risk factor. With EUR/USD only 5% from the upper barrier, small FX moves have outsized impact on valuation.

8. Conclusions & Recommendations

Z Group Perspective

- Product has become a **EUR 46M asset** from EUR 192M liability
- Gold well above strike provides intrinsic value protection
- EUR/USD strength toward 1.25 would trigger knockout and lock in gains
- *Risk: Severe gold crash + EUR weakness could still cause losses*

A Bank Perspective

- Facing **EUR 46M mark-to-market loss**
- ~95% probability of early knockout limits recovery potential
- Recent 7% gold correction helped, but insufficient
- *Options: Consider early termination negotiation*

Final Assessment: The structured gold forward has experienced a fundamental shift in economics. With ~95% knockout probability and average knockout time of 0.61 years, early termination is highly likely. The key risk factor is EUR/USD barrier proximity - any continuation of EUR strength would trigger termination and crystallize Z Group's gains.

GAAIF Challenge 2026 - Structured Gold Forward Pricing Analysis
Monte Carlo Simulation: 100,000 paths | Market Data: February 1, 2026